Ecological Network Structure Is Multidimensional: Rethinking the Stability Debate

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Abstract: Ecological networks are commonly characterised using a wide array of structural metrics, yet there remains little consensus on how these descriptors relate to ecological stability. Different studies frequently identify different structural predictors of stability, leading to long-standing debate about the role of network architecture in ecosystem dynamics. One possible explanation is that network structure is not a single property but a multidimensional feature of ecological systems. If so, commonly used metrics may capture different aspects of network organisation rather than alternative measures of the same underlying attribute. Here we test this idea by examining the relationships among thirty widely used food-web metrics across a compilation of empirical networks. We show that these descriptors organise into statistically robust modules that correspond to distinct dimensions of network architecture. Multivariate analyses further reveal that food webs occupy a low-dimensional structural space despite the large number of available metrics. Many commonly used descriptors capture overlapping structural information, whereas others represent independent axes of variation in how trophic interactions are organised. These results suggest that apparent inconsistencies in the structure–stability literature may arise because different studies measure different dimensions of network structure. Rather than asking which single metric predicts stability, analyses of ecological networks may benefit from explicitly considering the multidimensional nature of network architecture. By identifying the major structural domains underlying food-web organisation, this study provides a conceptual framework for interpreting network metrics and for more systematically linking network structure to ecological dynamics.

Keywords: food web, structure, dimensionality reduction

1 Introduction

Which metrics best represent the distinct stages of the energy-flow hierarchy, and do they capture different components of stability?

Food web structure self-organises into process-level modules, and those modules govern different components of stability.

Food web architecture is organised into emergent structural process modules, and different modules regulate different components of ecological stability.

Food web metrics self-organise into structural process modules, these modules define network space, and different modules regulate different stability components.

Should we be looking at if different metrics scale predictably with each other?

Emergent property (thinking like the cool tensor work/dynamic models) vs static metrics (using one measure of structure)

Ecological networks are commonly characterised using a large and diverse set of structural metrics, yet there remains little consensus on how these metrics should be interpreted or compared across studies (Vermaat et al. 2009; Lau et al. 2017). Measures of connectance, trophic level, modularity, omnivory, motif structure, and many others are routinely invoked as indicators of ecosystem organisation or stability. However, these descriptors often covary strongly, capture overlapping information, or reflect different scales of network organisation. As a result, studies using different subsets of metrics frequently arrive at conflicting conclusions about the relationship between network structure and ecological stability.

Ecological theory has long sought general principles linking the architecture of interaction networks to ecosystem stability. Early theoretical work framed this relationship in terms of system complexity, most famously through the analysis of random community matrices by Robert May, who showed that increasing species richness, interaction density, and interaction strength variance should reduce local dynamical stability in large ecological systems (May 1972, 1974). Subsequent work refined this prediction by demonstrating that specific forms of network organisation such as interaction structure, trophic hierarchy, and compartmentalisation can substantially alter stability outcomes (McCann 2000; Stouffer and Bascompte 2010; Allesina and Tang 2012). More recently, structural theories such as trophic coherence have proposed that food webs occupy constrained regions of structural space in which particular architectural features strongly influence system stability (Johnson et al. 2014). At the same time, ecological stability itself is now widely recognised as a multidimensional concept encompassing distinct dynamical properties such as persistence, resistance to

disturbance, and recovery dynamics (Tilman 1996; Ives and Carpenter 2007; Loreau and de Mazancourt 2013). Together these developments suggest that the relationship between network structure and ecological stability may be more complex than originally assumed, specifically that both structure and stability are multidimensional properties of ecological systems. If so, apparent inconsistencies in the structure–stability literature may arise not because theoretical predictions are incorrect, but because different studies measure different aspects of network architecture and different components of ecological stability.

One explanation for this inconsistency is that network structure is fundamentally multidimensional. Rather than representing a single property of ecosystems, structural metrics may capture different aspects of how interactions are organised and how energy moves through ecological systems. If this is the case, then individual metrics should not be expected to provide a universal predictor of stability. Instead, they may represent distinct dimensions of structural organisation that relate to different ecological mechanisms. Despite this possibility, most studies treat structural descriptors independently or select a small subset based on convention or data availability. Relatively few attempts have been made to examine whether commonly used network metrics organise into a smaller set of coherent structural dimensions that define the underlying space of ecological network architecture.

Ecological stability is likewise not a single property but a collection of related dynamical behaviours (Pimm 1984; Ives and Carpenter 2007). Different studies have emphasised different stability components, including the persistence of species [REF], the resistance of communities to disturbance [REF], and the speed with which systems recover following perturbation [REF]. These processes reflect distinct aspects of ecosystem dynamics and may depend on different features of network structure. From an energy-flow perspective, three stability mechanisms are particularly relevant: persistence, resistance, and return. Persistence refers to the continued maintenance of energy flow and species presence following species loss or environmental change. Resistance describes the degree to which disturbances propagate through ecological interactions, potentially amplifying or dampening perturbations. Return captures the capacity of a system to reorganise and re-establish stable interaction patterns following disturbance. Because these mechanisms operate through different ecological processes, they may be expected to depend on different aspects of network architecture. Structural features related to species roles and redundancy may influence persistence, whereas properties governing interaction pathways and network organisation may affect disturbance propagation or recovery dynamics.

Within this framework, metrics that are often treated as competing predictors of 'stability' instead emerge as complementary descriptors of different stability mechanisms (Thompson et al. 2012). Node-level metrics primarily relate to persistence, path-level metrics to resistance, and global organisational metrics to return dynamics. Some descriptors span multiple scales, reflecting the coupling between structural organisation

and emergent behaviour (Allesina and Tang 2012). This perspective provides a mechanistic explanation for why studies using different network metrics frequently report contrasting structure–stability relationships. Rather than reflecting inconsistency or redundancy, these differences arise because different metrics implicitly target different components of stability (Lau et al. 2017). By explicitly linking structural scale, energy flow, and stability mechanism, this framework provides a principled basis for interpreting network metrics and for selecting descriptors that align with specific ecological questions.

Here we investigate whether the large number of structural metrics used to characterise food webs can be reduced to a smaller set of coherent structural dimensions, and whether these dimensions relate to different components of ecological stability. Using a compilation of empirical food webs, we quantified a broad suite of commonly used network descriptors spanning species roles, interaction pathways, and global network organisation. We first examined whether these metrics self-organise into statistically robust modules based on their covariation across ecosystems. We then evaluated how these modules align with dominant multivariate axes of structural variation. Finally, we compared alternative representations of network structure to assess how different structural dimensions relate to three stability properties: robustness to species loss, spectral radius, and structural complexity. By explicitly mapping the multivariate structure of network architecture to stability outcomes, this analysis provides a principled framework for interpreting network metrics and for selecting structural descriptors that match specific ecological questions.

2 Materials & Methods

2.1 Data Compilation

We compiled food web data from Mangal (Poisot et al. 2016), Web of Life (Fortuna et al. 2014), and the canonical networks used by Vermaat et al. (2009), resulting in a total of XX networks. All networks were treated as binary and characterised using a suite of XX structural metrics (see Table 1).

Table 1: An informative caption about the different network properties. We use a combination of metrics from both the original Vermaat et al. (2009) paper as well as including those that have been identified by Thompson et al. (2012) and have been linked to emerging ecosystem properties such as stability

Label	Definition	Structural interpretation	Reference
Basal	Proportion of taxa with zero vulnerability (no consumers).	Quantifies the proportion of species representing basal energy inputs to the network.	
Top	Proportion of taxa with zero generality (no resources).	Describes the relative prevalence of terminal consumers in the network.	
Intermediate	Proportion of taxa with both consumers and resources.	Captures the proportion of species participating in both upward and downward energy transfer.	
Richness (S)	Number of taxa (nodes) in the network.	Describes network size.	
Links (L)	Total number of trophic interactions (edges).	Describes interaction density independent of network size.	
Connectance	L/S^2, where S is the number of species and L the number of links	Measures the proportion of realised interactions relative to all possible interactions.	Dunne et al. 2002
L/S	Mean number of links per species.	Captures average interaction density per taxon.	
Cannibal	Proportion of taxa with self-loops.	Quantifies the prevalence of cannibalistic interactions.	

Label	Definition	Structural interpretation	Reference
Herbivore	Proportion of taxa feeding exclusively on basal species.	Describes the representation of primary consumers.	
Intermediate	Percentage of intermediate taxa (with both consumers and resources)		
Trophic level (TL)	Prey-weighted trophic level averaged across taxa.	Captures the vertical organisation of energy transfer.	Williams and Martinez (2004)
MaxSim	Mean maximum trophic similarity of each taxon to all others.	Quantifies functional similarity based on shared predators and prey.	Yodzis and Winemiller (1999)
Centrality	Node centrality averaged across taxa (definition-dependent).	Captures the distribution of influence or connectivity among species.	Estrada and Bodin (2008)
ChLen	Mean length of all food chains from basal to top taxa.	Describes the average number of steps in energy-transfer pathways.	
ChSD	Standard deviation of food chain length.	Captures variability in pathway lengths.	
ChNum	Log-transformed number of distinct food chains.	Quantifies the multiplicity of alternative energy pathways.	
Path	Mean shortest path length between all species pairs.	Describes the average distance between taxa within the network.	
Diameter	Maximum shortest path length between any two taxa.	Captures the largest network distance between species.	

Label	Definition	Structural interpretation	Reference
Omnivory	Proportion of taxa feeding on resources at multiple trophic levels.	Describes vertical coupling of energy channels.	McCann (2000)
Loop	Proportion of taxa involved in trophic loops.	Quantifies the prevalence of cyclic interaction pathways.	
Prey:Predator	Ratio of prey taxa (basal + intermediate) to predator taxa (intermediate + top).	Describes the overall shape of the trophic structure.	
Diameter	Diameter can also be measured as the average of the distances between each pair of nodes in the network		
Clust	Mean clustering coefficient.	Measures the tendency for taxa sharing interaction partners to also interact with each other.	Watts and Strogatz (1998)
GenSD	Normalised standard deviation of generality.	Captures heterogeneity in the number of resources per taxon.	Williams and Martinez (2004)
VulSD	Normalised standard deviation of vulnerability.	Captures heterogeneity in the number of consumers per taxon.	Williams and Martinez (2004)
LinkSD	Normalised standard deviation of total links per taxon.	Quantifies variation in species connectivity.	
Intervality	Degree to which taxa can be ordered along a single niche dimension.	Measures the extent of niche ordering in trophic interactions.	Stouffer et al. (2006)

Label	Definition	Structural interpretation	Reference
S1 (Linear chain)	Frequency of three-node linear chains (A → B → C) with no additional links.	Captures the prevalence of simple, unbranched energy-transfer pathways.	Stouffer et al. (2007) Milo et al. (2002)
S2 (Omnivory)	Frequency of three-node motifs forming a feed-forward loop (A → B → C, A → C).	Describes vertical coupling of trophic levels within small subnetworks.	Stouffer et al. (2007) Milo et al. (2002)
S4 (Apparent competition)	Frequency of motifs where one consumer feeds on two resources (A → B ← C).	Captures the prevalence of shared-predator structures among resources.	Stouffer et al. (2007) Milo et al. (2002)
S5 (Direct competition)	Frequency of motifs where two consumers share a single resource (A ← B → C).	Describes the occurrence of shared-resource structures among consumers.	Stouffer et al. (2007) Milo et al. (2002)
trophicVar	Measure of how much the trophic positions of species deviate from the mean trophic level.	Variance is linked to chain length, with low variance indicating few long chains and high variance indicates long chains and varied interactions	Pimm (1982)
TrophicCoherence	Measure how well the species in a food web fit into discrete trophic levels.	Highly coherent, neatly layered food webs are more stable to perturbations	Johnson et al. (2014)

In addition to network structural descriptors, we quantified multiple complementary measures of ecological stability that reflect different dynamical processes. Specifically, we included (1) Robustness (R_{50}), which measures the proportion of secondary extinctions following sequential species loss and captures tolerance to perturbation (Jonsson et al. 2015); (2) Complexity (SVD), defined as the Shannon entropy of the singular

value decomposition of the adjacency matrix, representing heterogeneity in interaction pathways and potential system-level fragility (Strydom et al. 2021); (3) Spectral Radius ( ), the largest real part of the eigenvalues of the adjacency matrix, providing a global indicator of dynamic stability related to interaction strength aggregation (Staniczenko et al. 2013); and (4) Structural Controllability, which evaluates the minimum number of driver species required to control the network's dynamics and thus captures the capacity for directed reorganisation following perturbation (Liu et al. 2011). Together, these metrics operationalise multiple facets of ecological stability, linking structural organisation to distinct dynamical processes within food webs.

2.2 Identification of Structural Modules

We tested the hypothesis that structural descriptors of food webs are organised into statistically coherent modules reflecting shared ecological function or scaling relationships, rather than forming arbitrary clusters driven by sampling noise. If such modular organisation exists, then metrics within a module should exhibit strong internal correlation relative to metrics assigned to different modules. The resulting clusters should be robust to resampling of the data. The identified modular structure should exceed expectations under null models of random association among metrics.

We quantified pairwise associations among the XX structural metrics using Pearson correlations computed across food webs. Because ecological metrics may covary either positively or negatively depending on scaling relationships, we constructed two alternative distance matrices: $1 - r$, which preserves the sign of correlations and distinguishes positive from negative association and $1- \mid r \mid$, which groups variables based on the magnitude of their association regardless of sign. These two definitions allow us to test whether modular structure depends on directional relationships or simply on the strength of coupling among metrics. Hierarchical clustering was performed using average linkage on each distance matrix to identify candidate structural modules.

The optimal number of clusters was evaluated across a range of partition sizes ($k = 2$–10) using average silhouette width to assess within-cluster cohesion and between-cluster separation. To further evaluate cluster robustness, we implemented bootstrap resampling with 1,000 bootstrap replicates. This procedure estimates approximately unbiased (AU) p-values for each cluster, quantifying the probability that a cluster is supported under repeated resampling of the data. Clusters were considered statistically robust when they exhibited high silhouette support, and approximately unbiased bootstrap support 0.95. This dual criterion ensures that identified modules are both structurally coherent and stable to sampling variation.

2.3 Multivariate Structure and Module–Axis Alignment

To evaluate whether structural metrics of food webs organise into coherent multivariate modules and whether those modules define principal axes of variation, we conducted a principal component analysis (PCA) followed by a permutation-based test of module–axis alignment. Under this framework, principal components define dominant structural gradients in the metric space, whereas modules represent hypothesised mechanistic groupings of structurally related metrics. Demonstrating alignment between these two representations would suggest that modular decomposition captures fundamental axes of ecological variation.

Skewed count-based metrics (*e.g.,* link counts, interval counts, and related quantities) were log-transformed using $log(x + 1)$ to reduce right skew. All remaining variables were standardised to zero mean and unit variance prior to analysis to ensure that metrics measured on different scales contributed equally to the ordination.

To quantify how structural modules align with principal axes, we decomposed variance in each principal component according to module membership. For each module m and principal component k, we computed:

$$A_{mk} = \sum_{i \in m} l_{ik}^2$$

where l_{ik} is the loading of metric i on PC_k and A_{mk} represents the fraction of variance in PC_k attributable to metrics in module m. Because squared loadings sum to one within each principal component, this provides a direct partitioning of PC variance across modules. This produces a module $\times$ PC matrix describing geometric alignment between modular structure and multivariate axes.

To evaluate whether observed module–PC alignment exceeded expectations under random module structure, we implemented a permutation test. We randomly permuted module labels among metrics 1 000 times while preserving the number and size distribution of modules, and the PCA loadings. For each permutation r, we recomputed $A_{mk}^{(r)}$ so as to form a null distribution for each module–PC pair. From this we computed p-values and corresponding z-scores, which was then used to infer significant alignment when $p_{mk} < 0.05 \wedge |Z_{mk}| > 1.96$ where:

$$p_{mk} = P(A_{mk}^{(r)} > A_{mk})$$

and:

$$Z_{mk} = \frac{A_{mk} - \mu_{mk}}{\sigma_{mk}}$$

Additionally we also evaluated if modules collectively aligned with PCA structure beyond random expectation. Overall concentration of variance within module–PC space was calculated as $T = \sum_{m,k} A_{mk}^2$ and significance was determined by comparing the observed (T) the permutation distribution ($T^{(r)}$). Where $p_{global} = P(T^{(r)} \geq T)$. A significant result indicates that module structure explains multivariate variance better than random groupings.

2.4 Structural Metric Reduction and Representation of Network Space

To evaluate how network structure predicts ecological stability, we first reduced a high-dimensional set of topological metrics into alternative, biologically interpretable predictor sets. Because dimensionality reduction can fundamentally shape inference, we explicitly compared two conceptually distinct representations of network structure: (1) structural domain representatives and (2) latent axes of network variation.

2.4.1 Identification of Structural Domains

We previously grouped network metrics into clusters based on their pairwise correlations, such that each cluster represented a structurally coherent domain (*e.g.,* trophic composition, centralisation, path structure). Clustering was performed using correlation-based distance, ensuring that metrics grouped together reflected shared structural information rather than raw scale. To construct a reduced predictor set from these domains, we selected a single representative metric per cluster using a medoid approach. Within each cluster, we computed the absolute correlation matrix among member metrics and defined distance as $1 - \mid r \mid$. The medoid was identified as the metric minimising mean pairwise distance to other metrics in the cluster. This approach preserves structural diversity while minimising redundancy and does not rely on principal component loadings, which can bias selection toward dominant variance axes. The resulting 'cluster medoid' set retained one interpretable metric per structural domain. Importantly, this procedure prioritises preservation of structural domains rather than overall variance magnitude, allowing low-variance but potentially mechanistically relevant features of network topology to be retained.

2.4.2 Extraction of Latent Structural Axes

As a complementary representation of network structure, we performed principal component analysis (PCA) on the scaled full metric matrix. Principal components were retained until cumulative explained variance exceeded 80%, yielding a set of orthogonal axes describing the dominant gradients of variation in network topology.

These retained PC scores constitute a low-dimensional latent representation of network space. Unlike cluster medoids, which preserve structural categories, PC scores preserve maximal variance and ensure orthogonality among predictors. We also explored a third reduction approach—selecting the single highest-loading metric per retained PC. However, because principal components are linear combinations of multiple metrics, this PC-dominant metric strategy substantially reduces the variance represented by each axis. Consequently, this approach captures less of the total structural variance.

2.4.3 Conceptual Contrast Between Representations

The cluster-medoid and PCA-score representations reflect different theoretical assumptions about how structure relates to stability. The cluster-medoid approach assumes that ecological stability responds to discrete structural domains that may vary independently and need not align with the dominant gradients of network variation. In contrast, the PCA-score approach assumes that stability responds primarily to the major axes of variation in network topology, regardless of their interpretability in terms of structural domains. By explicitly comparing predictive performance across these representations, we test whether ecological stability is better explained by domain-specific structural mechanisms or by global geometric gradients of network organisation.

2.4.4 Structural Control of Stability

To evaluate how hierarchical representations of network structure explain variation in stability, we modelled four complementary stability outcomes: Robustness (R_{50}), Complexity (SVD), Spectral Radius ( ), and Structural Controllability. These measures capture distinct dynamical processes governing ecosystem persistence, resistance, and controllability. Elastic net regression models were fit separately for each stability metric using three structural representations (cluster medoids, PC-dominant metrics, and full PCA scores), allowing us to compare the structural determinants of different stability processes. All predictors and response variables were standardised prior to analysis to allow direct comparison of coefficient magnitudes.

Elastic net regularisation was chosen because it permits inference under correlated predictors by interpolating between ridge (distributed shrinkage) and lasso (sparse selection) penalties. The mixing parameter α (0–1) determines the degree of sparsity, with lower α values indicate distributed influence across many correlated structural descriptors (ridge-like behaviour), whereas higher α values indicate that stability is driven by a smaller subset of structural predictors (lasso-like sparsity)..

2.4.4.1 Model Evaluation

Predictive performance was assessed using repeated v-fold cross-validation (5 folds x 10 repeats). Within each training partition, models were fit across a grid of values (0–1 in increments of 0.25), and the optimal penalty strength ( ) was selected via internal cross-validation within each training partition. Performance was quantified as cross-validated R^2 computed on held-out test partitions. After cross-validation, final models were refit to the full dataset using the mean selected α, and standardised coefficients were extracted to characterise the direction and magnitude of structural effects. The objective of this analysis was not predictive optimisation, but rather to quantify how structural organisation governs different stability processes. The elastic net mixing parameter also provides insight into the architecture of structural control. To quantify the relative contribution of different structural modules, standardized regression coefficients from the final models were squared and aggregated within modules. The proportion of explained variance attributable to each module was then scaled by the model's cross-validated R^2, providing a measure of the absolute variance in stability explained by each structural component.

3 Results

3.1 Structural Metrics Organise into Robust Modules

Hierarchical clustering of structural descriptors revealed clear modular organisation within food-web architecture Figure 1. Silhouette analysis indicated an optimal partition of the signed correlation matrix at k=7 modules, with a maximum average silhouette width of [insert value]. Clustering based on absolute correlations produced a slightly coarser but comparable solution (k=5), demonstrating that the modular structure is robust to the treatment of correlation sign and reflects strong underlying association patterns among metrics.

[Figure 1 about here.]

Bootstrap resampling further supported the stability of the major clusters. Most principal modules exhibited high approximately unbiased (AU) support values (AU [insert value]), indicating that the identified clusters are not artefacts of sampling variability but represent statistically robust groupings of structural descriptors.

The resulting modular partition identified coherent groups of metrics describing distinct aspects of network architecture Table 2. These included a large-scale architectural module integrating size and motif-based descriptors, modules capturing trophic organisation and energy routing, and two modules consisting of isolated structural descriptors that function independently from broader metric groupings. The six-module solution Table 2 reveals a hierarchical decomposition of food-web structural space into interpretable ecological dimensions. Collectively, these modules illustrate that network organization is shaped by multiple statistically supported axes of structural variation, reflecting size scaling, integration, energy flow, and control asymmetry.

Table 2: Structural Modules Identified From Hierarchical Clustering.

Module Name	Metrics Included	Ecological Interpretation
Macro Complexity	richness, links, l_S, S4, S5, intervals	Captures large-scale expansion of network structure and combinatorial complexity associated with increasing system size
Trophic Integration	connectance, diameter, intermediate, omnivory, cannibal, TL, ChLen, ChSD, S1, S2, loops, Clust, trophicCoherence, trophicVar	Describes how ordered the trophic structure is and how tightly species are embedded across trophic levels
Energy Transport	mean distance, path length	Quantifies geometric structure of energy routing independent of network size or trophic structure
Trophic Role Asymmetry	basal, predpreyRatio, GenSD, LinkSD, MaxSim	Captures structural imbalance across trophic roles, but it also incorporates heterogeneity in consumer behaviour and redundancy.
Control Heterogeneity	top, VulSD, ChNum	Reflects heterogeneity in top-down regulation and predation pressure distribution
Basal Control	herbivory, centrality	Captures energy entry points and dominant structural hubs

3.2 Structural Modules Align with Dominant Axes of Network Variation

Principal component analysis of standardised structural metrics revealed strong dimensional compression in food-web structure Figure 2. The first three principal components accounted for 68.5% of total variance (PC1 = 31.2%, PC2 = 21.5%, PC3 = 15.6%), with 80.7% of variance explained by the first five components. Subsequent axes contributed progressively smaller proportions of variance, indicating that a small number of dominant gradients summarise most structural variation.

[Figure 2 about here.]

[Figure 3 about here.]

Inspection of the variance explained by module-level contribution Figure 4 indicate that PC1 primarily reflects variation in Macro-Architectural Complexity and Trophic Integration & Structural Consistency, capturing differences in network size, link density, and intermediate-level trophic ordering. PC2 is influenced largely by Control Heterogeneity and Trophic Integration, representing variation in top-down control and cross-level integration. PC3 contrasts Energy Transport Geometry with Control Heterogeneity, reflecting independent variation in energy routing and uneven distribution of interaction pressure. These results suggest that food-web structure can be summarised by a small number of ecologically interpretable axes, representing orthogonal gradients of size and complexity, trophic integration, and control heterogeneity.

[Figure 4 about here.]

The global permutation test indicated significant alignment between the modular partition and the principal component structure ($p = 0.05$), demonstrating that the observed module assignment explains the multivariate geometry of structural metrics better than randomly assigned modules. Alignment strength varied substantially across modules Figure 5. Macro-Architectural Complexity (Module 1) exhibited the strongest and most extensive alignment, showing significant associations with multiple principal components and a dominant loading on PC1, indicating that it contributes substantially to the primary structural gradient associated with network size, link density, and motif diversity. Trophic Integration & Structural Consistency (Module 6) displayed broad alignment across three principal components, with its strongest association on PC1, reflecting its contribution to intermediate-level ordering and consistency in trophic level structure. Integration & Dominance (Module 2) and Control Heterogeneity & Top-Down Variability (Module 5) showed more limited alignment, each significantly associated with one to two principal components, highlighting variation in network cohesion, dominance concentration, and top-down pressure that maps onto secondary structural gradients. Energy Transport Geometry (Module 3) was significantly aligned with PC3, indicating that energy routing structure represents an axis largely orthogonal to the dominant size and integration gradients. Trophic Asymmetry (Module 4) showed modest alignment with higher-order components, suggesting that skew between basal and consumer dominance captures subtler variation in network structure. Overall, significant module–PC associations were concentrated in a subset of principal components, reinforcing the interpretation that modular structure maps onto specific, ecologically interpretable structural gradients rather than uniformly spanning the full multivariate space.

[Figure 5 about here.]

Collectively, these results demonstrate that food-web structural metrics exhibit strong low-dimensional

organisation and that independently identified structural modules are non-randomly aligned with the dominant axes of variation. The significant global alignment supports the interpretation that modular decomposition reflects meaningful geometric structure rather than clustering artifacts. The broad influence of Macro-Architectural Complexity (Module 1) and Trophic Integration & Structural Consistency (Module 6) across multiple principal components suggests a hierarchical organisation, in which these core modules govern the primary gradients of variation, while additional modules—including Integration & Dominance, Control Heterogeneity, Energy Transport Geometry, and Trophic Asymmetry—capture secondary and more specialised structural dimensions. This pattern indicates that food web structure is shaped by a combination of network size, intermediate-level trophic ordering, energy flow geometry, and top-down control, all of which contribute to distinct, interpretable axes of variation.

3.3 Predictors of stability

3.3.1 Dimensional Reduction Characterisation

Representation	Dimensionality	Variance Preserved	Structure Type
Medoids	7	23%	Domain sampling
PC-dominant metrics	4	13%	Axis proxy sampling
PC scores	5	80%	True latent space

To evaluate how alternative structural representations differed in their information content and redundancy, we quantified variance retention, internal correlation structure, effective dimensionality, and geometric similarity among representations.

Variance Retention: The three representations differed substantially in the proportion of total structural variance preserved (Fig. S1A). The cluster-medoid representation retained ~23% of the total variance in the full metric space despite reducing dimensionality to seven predictors. In contrast, the PC-dominant metric set retained only ~13% of total variance, reflecting the fact that individual metrics capture only a fraction of each principal component's multivariate structure. As expected by construction, the retained PC-score representation preserved approximately 80% of total variance. Thus, domain-based reduction preserved more distributed structural information than selecting one metric per PC axis, whereas the PC-score representation maximally preserved dominant gradients of network variation.

Internal Redundancy: Representations also differed in their internal correlation structure (Fig. S1B). PC scores were orthogonal by definition (mean |r| 0), indicating complete statistical independence among

predictors. In contrast, cluster medoids exhibited moderate residual correlation (mean |r| = X), suggesting partial overlap among structural domains. PC-dominant metrics showed comparable (or higher/lower — insert result) redundancy relative to medoids. These differences indicate that the three approaches vary not only in information retention but also in predictor independence.

Effective Dimensionality: We next quantified effective dimensionality as the number of axes required to explain 80% of variance within each reduced predictor set (Fig. S1C). The PC-score representation required X axes, reflecting its design to capture dominant structural gradients. The cluster-medoid representation required X axes to reach the same threshold, indicating that despite containing seven predictors, structural variation was concentrated along fewer effective dimensions. The PC-dominant set exhibited the lowest effective dimensionality (X axes), consistent with its reduced variance retention. Together, these results show that dimensional compression differed across approaches not only in magnitude but in the distribution of variance across axes.

Overall, the three structural representations differed substantially in variance retention, redundancy, and effective dimensionality. Cluster medoids preserved moderate variance while maintaining domain interpretability, PC-dominant metrics retained minimal total variance, and PC scores preserved dominant structural gradients while ensuring predictor orthogonality. These differences establish that the representations encode distinct aspects of network topology, justifying their empirical comparison in predictive analyses of stability.

[Figure 6 about here.]

3.3.2 Predictive Performance Across Structural Representations

3.3.2.1 Predictor Importance and Structural Drivers

Across all stability components, elastic net regularisation revealed that structural predictors contributed differently depending on how network space was represented. The degree of regularisation (α, Figure 7) and the distribution of non-zero coefficients Figure 8 varied systematically, indicating that structural abstraction alters which aspects of topology emerge as dominant drivers. Models based on fine-scale Cluster Dominant descriptors often reflected Distributed Control across correlated topological features, seen most clearly in the complexity ($\alpha = 0$) and control ($\alpha = 0$) models. In these instances, structural heterogeneity and connectivity descriptors jointly contributed to stability. By contrast, PC Dominant models tended to concentrate predictive weight onto a smaller subset of variables, frequently resulting in a Sparse architecture ($\alpha = 1$ for robustness, , and complexity). This suggests that dimensional compression isolates key structural bottlenecks, such as predpreyRatio or intermediate taxa proportions, which summarise broader variation. When modelled using

PC Scores, structural influence collapsed onto dominant components, with PC1 or PC3 acting as the primary axes of influence.

[Figure 7 about here.]

[Figure 8 about here.]

3.3.2.2 Structural Drivers of Stability

Predictive performance (R^2) differed substantially across stability components Figure 9, revealing how strongly each facet is encoded in static network architecture. Across all representations, and control exhibited the strongest structural determination. reached its highest cross-validated predictive performance ($R^2 = 0.723$) under the multivariate PC Score representation, driven primarily by PC1 (estimate = 0.653). Control showed even higher determination in the Cluster Dominant model ($R^2 = 0.760$), where GenSD emerged as a key distributed driver.

Complexity showed intermediate structural predictability ($R^2 = 0.248$), performing best under PC Score abstraction. Conversely, Robustness exhibited the weakest structural predictability across all representations ($R^2_{max} = 0.181$). This pattern implies that robustness is not strongly determined by static topology alone and may depend on dynamical or context-dependent processes. Notably, the shift in R^2 across representations—such as jumping from 0.262 in the PC Dominant model to 0.723 in the PC Score model—indicates that the apparent strength of structural control is contingent on the chosen structural lens.

[Figure 9 about here.]

3.3.3 Differential Effects Across Stability Components

The relationship between network structure and stability was governed by different structural hierarchies. Complexity was characterised by a distributed or intermediate architecture in most representations, reflecting the cumulative effect of interacting features. In contrast, displayed a strong alignment with coordinated multivariate axes, with the Trophic Integration module explaining a significant portion (30.2%) of the variance in the Cluster Dominant model.

Control exhibited a uniquely representation-sensitive architecture: while the Cluster Dominant model suggested a Distributed regime ($\alpha = 0$), the PC Score model favoured a Sparse regime (=1). This indicates that while the structural signals for control are widespread (involving Trophic Role Asymmetry), they can be distilled into a single, high-leverage structural axis (PC1, estimate = -0.791). Finally, Robustness shifted towards Sparsity under PC abstraction, but its low R^2 suggests that even its bottleneck structural drivers (such as

PC3) are only weak constraints on the system's ability to withstand species loss.

Collectively, these results demonstrate that stability components occupy distinct positions within structural space. Complexity is shaped by distributed fine-scale topology, emerges from coordinated multivariate gradients, and robustness remains weakly constrained by static architecture. Rather than reflecting a single structural driver, ecosystem stability is organised along layered, facet-specific dimensions of network architecture.

[Figure 10 about here.]

4 Discussion

4.1 Network structure is multidimensional

For several decades, ecological theory has sought to understand how the structure of food webs influences ecological stability. Yet despite extensive empirical and theoretical work, consensus has remained elusive. Different studies frequently identify different structural predictors of stability, and results often appear contradictory. Connectance has been linked both positively and negatively to stability [REF], omnivory has been argued to stabilise or destabilise communities [REF], and measures of modularity, trophic organisation, or motif structure have each been proposed as key determinants of ecosystem dynamics [REF].

Our results suggest that much of this inconsistency may arise from a conceptual simplification - network structure is rarely a single property. We demonstrate that food web structure exhibits strong modular organisation among commonly used network metrics Figure 1, with structural descriptors grouped into seven statistically supported modules spanning macro-architectural complexity, trophic integration, routing geometry, and trophic imbalance. Instead, food-web architecture appears to be organised along several coherent dimensions of variation, with principal component analysis revealing that this high-dimensional set of metrics collapses onto a small number of dominant structural gradients Figure 2, Figure 3, Figure 4. Importantly, the independently identified structural modules align non-randomly with these principal gradients, indicating that the modular decomposition captures real axes of ecological variation rather than statistical artifacts. This suggests that the structural metrics commonly used in ecological studies do not behave as independent descriptors, nor do they represent interchangeable proxies for a single structural attribute, but rather that they cluster into statistically robust modules that capture distinct aspects of network organisation.

This modular structure implies that ecological networks occupy a multidimensional structural space, in which different metrics describe different axes of variation in how interactions are arranged. In this view, commonly

used descriptors such as connectance, trophic level, motif frequencies, or network distance are not competing measurements of the same property but rather partial projections of a more complex architectural system. The structure–stability debate has therefore often compared results derived from fundamentally different structural dimensions.

4.2 Structural modules reveal dimensions of network architecture

The clustering analysis revealed several consistent structural domains within food-web architecture Table 2. The largest module captured macro-architectural expansion, combining species richness, interaction density, motif frequencies, and intervality. Because these metrics scale together as networks grow, this module reflects the combinatorial growth of interaction space with increasing species richness. In contrast, trophic integration metrics—including connectance, omnivory, trophic level, and clustering—formed a distinct module capturing how tightly energy pathways are embedded across trophic levels.

The emergence of these modules indicates that food-web structure is not best understood as a flat collection of descriptors but as a hierarchically organised architectural system. Certain structural properties scale together as networks expand, while others vary independently and represent alternative ways of organising trophic interactions.

Importantly, the existence of these modules suggests that the large number of metrics currently used in network ecology largely represent a smaller number of underlying structural domains. Many descriptors capture overlapping information, whereas others represent distinct aspects of network organisation that are rarely considered together. Recognising these domains provides a principled basis for reducing metric redundancy and for interpreting structural patterns across ecosystems.

4.3 Structural domains and the mechanisms of stability

The modular organisation of structural metrics identified in our analysis provides a framework for linking distinct ecological processes to different components of ecosystem stability. Because each module captures a different aspect of network architecture, they likely influence stability through different mechanistic pathways. Interpreting the modules in light of existing food web theory suggests that multiple ecological processes such as community assembly, trophic integration, energetic constraints, and niche overlap jointly shape how perturbations propagate through ecological networks.

The module associated with macro-architectural complexity reflects the structural consequences of increasing community size and interaction opportunities. Metrics describing species richness, interaction density, motif

diversity, and large-scale path structure covary strongly, indicating that these properties scale together as food webs accumulate species. Classic theoretical work predicted that increasing complexity should destabilise ecological systems by increasing the density of interactions within the community matrix (May 1972). However, subsequent studies demonstrated that the stability consequences of complexity depend strongly on how interactions are distributed across the network (McCann et al. 1998; Montoya et al. 2006). In this context, the macro-architectural module may primarily influence stability through its effects on the overall dimensionality of species interactions and the number of pathways through which perturbations can spread.

A second structural domain describing trophic integration includes connectance, omnivory, trophic level, and clustering. These metrics capture how densely interactions couple trophic levels and the extent to which energy channels are interconnected across the food web. Theoretical work has long suggested that omnivory and cross-level feeding interactions can alter system stability by redistributing interaction strengths and dampening oscillatory dynamics (Polis and Strong 1996; McCann and Hastings 1997). Empirical analyses of interaction strengths similarly show that food webs often contain many weak links embedded within more complex interaction structures, a pattern that can stabilise population dynamics despite high connectance (McCann et al. 1998; Emmerson and Yearsley 2004). Variation in this structural domain therefore likely influences stability through the degree of coupling among trophic pathways and the distribution of interaction strengths across the network.

A third module describes the geometry of energy transport through the network, capturing variation in path length distributions and food-chain structure. The independence of these metrics from those describing network size suggests that the organization of trophic pathways represents a distinct structural dimension. Recent work has emphasised the importance of trophic pathway organisation in determining the stability of ecological networks. In particular, measures of trophic coherence and pathway structure have been shown to strongly influence the propagation of perturbations and the stability of dynamical equilibria (Johnson et al. 2014). Differences in pathway geometry may therefore affect how disturbances move through the network, influencing the amplification or attenuation of population fluctuations along trophic chains.

Other structural modules reflect variation in trophic role distributions and interaction heterogeneity across species. Metrics describing basal species prevalence and predator–prey ratios capture trophic asymmetry within the network, reflecting the energetic pyramid structure that arises from constraints on energy transfer across trophic levels. Such asymmetries influence both the distribution of interaction strengths and the sensitivity of higher trophic levels to perturbations originating at lower levels of the food web (Polis and Strong 1996). In contrast, modules describing heterogeneity in predation pressure capture how evenly consumer

impacts are distributed among species, which may influence the likelihood of cascading effects following perturbations.

Finally, descriptors of trophic similarity form an independent structural dimension associated with functional redundancy among species. Ecological theory predicts that redundancy in species roles can enhance robustness to species loss because remaining species may partially compensate for lost interactions (Walker 1992). Variation in trophic similarity and shared resource use may therefore play a particularly important role in determining robustness to species removal and the potential for extinction cascades (Montoya et al. 2006).

Taken together, these relationships suggest that different components of ecological stability may be governed by distinct domains of network architecture. Structural properties associated with community size and trophic integration may primarily influence equilibrium stability, whereas pathway geometry may shape dynamical variability by determining how perturbations propagate through trophic interactions. At the same time, heterogeneity in species roles and trophic similarity may influence robustness to species loss by determining the degree of functional redundancy within the network. Recognising these distinct structural domains helps clarify why previous studies have reported contrasting relationships between network structure and stability: different analyses may emphasise different aspects of network architecture, each linked to separate ecological mechanisms.

4.4 Reinterpreting the structure–stability debate

The multidimensional nature of network structure has important implications for how stability relationships are interpreted. Much of the historical debate surrounding ecological stability has implicitly treated structure as a single explanatory variable. Studies typically select one or a small number of metrics and evaluate their relationship with a particular stability measure. However, if structural descriptors capture different architectural dimensions, then such analyses effectively examine different slices of structural space.

Our stability analyses support this multidimensional interpretation. Different stability components responded to different structural representations and levels of abstraction. Spectral radius (ρ) exhibited the strongest structural determination and was best predicted when network structure was expressed as multivariate principal component axes. In contrast, complexity was most predictable from fine-scale structural descriptors, indicating distributed influence across multiple topological features. Robustness to species loss showed comparatively weak structural determination across representations.

These results help reconcile apparently conflicting conclusions in the historical structure–stability literature. Theoretical work emphasising stability, such as that of May [REF], effectively captures structural variation

along global connectivity gradients. In contrast, studies emphasising trophic pathways and energy dampening—such as those following McCann [REF] may instead capture variation in routing geometry or trophic integration. Because these architectural domains vary independently in empirical networks, different studies may observe contrasting stability relationships even when examining the same ecosystems.

Under this interpretation, apparently conflicting results across studies become less surprising. A study focusing on connectance may capture variation in trophic integration, while another emphasising motif frequencies may instead reflect changes in network expansion and pathway complexity. Each metric may relate to stability through different mechanisms, yet comparisons between them have often been framed as competing tests of the same hypothesis. Rather than asking which structural metric best predicts stability, it may therefore be more productive to ask which dimension of network architecture is relevant to a given stability mechanism. Stability itself is a multidimensional property, encompassing processes such as persistence, resistance to disturbance, and the reorganisation of interaction pathways following perturbation. Different structural domains may influence these processes through distinct ecological mechanisms. Our analysis does not attempt to resolve which particular metrics control specific stability outcomes. Instead, it highlights the need to reframe the problem: understanding ecological stability requires considering how multiple structural dimensions interact with different dynamical processes.

4.5 Implications for the design of network analyses

Our comparison of alternative structural representations highlights the consequences of dimensional reduction choices. Domain-based medoid representations preserved moderate structural variance while maintaining ecological interpretability, whereas selecting a single metric per principal component retained only a small fraction of total structural information. By contrast, multivariate PC scores preserved most structural variance but abstracted away from direct ecological interpretation. These results indicate that methodological choices about how network structure is summarised can substantially alter inference about stability drivers.

Recognising the multidimensional structure of food webs has practical implications for ecological analysis. First, it suggests that the common practice of selecting individual network metrics in isolation may obscure important structural relationships. Because many metrics capture overlapping aspects of network architecture, analyses based on a single descriptor may inadvertently reflect broader structural domains. Second, dimensionality reduction approaches provide a useful framework for representing network structure more coherently. By identifying modules of related metrics or latent axes of structural variation, researchers can characterise the major architectural gradients that define network organisation. Such approaches preserve the diversity of structural features while avoiding redundancy among descriptors. Finally, this perspective encourages a shift

from metric-driven analyses toward process-driven structural representation. Rather than selecting metrics based on convention, analyses can instead focus on structural domains most relevant to the ecological process under investigation. This approach may provide clearer mechanistic insight into how network architecture shapes ecosystem dynamics.

4.6 Toward a structural framework for ecological networks

The results presented here suggest that the architecture of ecological networks can be understood as a low-dimensional structural system emerging from the interactions among species and their trophic roles. Within this system, individual metrics represent different projections of a broader architectural space. Recognising this structure does not eliminate the complexity of ecological networks, but it provides a way to organise and interpret that complexity more coherently. More broadly, this perspective highlights the importance of treating network structure as a multivariate ecological property rather than a collection of independent descriptors. Just as biodiversity is now understood through multiple complementary dimensions, including richness, composition, and functional traits, the structure of ecological networks may be best described through a set of interacting architectural domains. Future work linking network structure to ecosystem dynamics may therefore benefit from explicitly considering this multidimensional framework. By aligning structural dimensions with ecological processes and dynamical mechanisms, it may be possible to move beyond debates over individual metrics and toward a more unified understanding of how interaction networks shape the stability of ecological systems.

5 Conclusion

The long-standing structure–stability debate in ecology may not reflect disagreement about ecological mechanisms, but rather the fact that network structure itself is multidimensional, and different studies have examined different dimensions of the same architectural system.

References

Allesina, S., and S. Tang. 2012. Stability criteria for complex ecosystems. Nature 483:205–208.

Emmerson, M., and J. M. Yearsley. 2004. Weak Interactions, Omnivory and Emergent Food-Web Properties. Proceedings: Biological Sciences 271:397–405.

Estrada, E., and Ö. Bodin. 2008. Using Network Centrality Measures to Manage Landscape Connectivity. Ecological Applications 18:1810–1825.

Fortuna, M. A., R. Ortega, and J. Bascompte. 2014. The Web of Life.

Ives, A. R., and S. R. Carpenter. 2007. Stability and Diversity of Ecosystems. Science 317:58–62.

Johnson, S., V. Domínguez-García, L. Donetti, and M. A. Muñoz. 2014. Trophic coherence determines food-web stability. Proceedings of the National Academy of Sciences 111:17923–17928.

Jonsson, T., S. Berg, A. Pimenov, C. Palmer, and M. Emmerson. 2015. The reliability of R50 as a measure of vulnerability of food webs to sequential species deletions. Oikos 124:446–457.

Lau, M. K., S. R. Borrett, B. Baiser, N. J. Gotelli, and A. M. Ellison. 2017. Ecological network metrics: Opportunities for synthesis. Ecosphere 8:e01900.

Liu, Y.-Y., J.-J. Slotine, and A.-L. Barabási. 2011. Controllability of complex networks. Nature 473:167–173.

Loreau, M., and C. de Mazancourt. 2013. Biodiversity and ecosystem stability: A synthesis of underlying mechanisms. Ecology Letters 16:106–115.

May, R. M. 1972. Will a Large Complex System be Stable? Nature 238:413–414.

———. 1974. Stability and Complexity in Model Ecosystems (Vol. 1). Princeton University Press.

McCann, K. S. 2000. The diversity–stability debate. Nature 405:228–233.

McCann, K., and A. Hastings. 1997. Re-Evaluating the Omnivory-Stability Relationship in Food Webs. Proceedings: Biological Sciences 264:1249–1254.

McCann, K., A. Hastings, and G. R. Huxel. 1998. Weak trophic interactions and the balance of nature. Nature 395:794–798.

Milo, R., S. Shen-Orr, S. Itzkovitz, N. Kashtan, D. Chklovskii, and U. Alon. 2002. Network Motifs: Simple Building Blocks of Complex Networks. Science 298:824–827.

Montoya, J. M., S. L. Pimm, and R. V. Solé. 2006. Ecological networks and their fragility. Nature 442:259–264.

Pimm, S. L. 1982. Food Webs. Springer Netherlands, Dordrecht.

———. 1984. The complexity and stability of ecosystems. Nature 307:321–326.

Poisot, T., B. Baiser, J. Dunne, S. Kéfi, F. Massol, N. Mouquet, T. N. Romanuk, et al. 2016. Mangal – making ecological network analysis simple. Ecography 39:384–390.

Polis, G. A., and D. R. Strong. 1996. Food Web Complexity and Community Dynamics. The American Naturalist 147:813–846.

Staniczenko, P. P. A., J. C. Kopp, and S. Allesina. 2013. The ghost of nestedness in ecological networks. Nature Communications 4:1391.

Stouffer, D. B., and J. Bascompte. 2010. Understanding food-web persistence from local to global scales. Ecology Letters 13:154–161.

Stouffer, D. B., J. Camacho, and L. A. N. Amaral. 2006. A robust measure of food web intervality. Proceedings of the National Academy of Sciences 103:19015–19020.

Stouffer, D. B., J. Camacho, W. Jiang, and L. A. Nunes Amaral. 2007. Evidence for the existence of a robust pattern of prey selection in food webs. Proceedings of the Royal Society B: Biological Sciences 274:1931–1940.

Strydom, T., G. V. Dalla Riva, and T. Poisot. 2021. SVD Entropy Reveals the High Complexity of Ecological Networks. Frontiers in Ecology and Evolution 9.

Thompson, R. M., U. Brose, J. Dunne, R. O. Hall, S. Hladyz, R. L. Kitching, N. D. Martinez, et al. 2012. Food webs: Reconciling the structure and function of biodiversity. Trends in Ecology & Evolution 27:689–697.

Tilman, D. 1996. Biodiversity: Population Versus Ecosystem Stability. Ecology 77:350–363.

Vermaat, J. E., J. A. Dunne, and A. J. Gilbert. 2009. Major dimensions in food-web structure properties. Ecology 90:278–282.

Walker, B. H. 1992. Biodiversity and Ecological Redundancy. Conservation Biology 6:18–23.

Watts, D. J., and S. H. Strogatz. 1998. Collective dynamics of "small-world" networks. Nature 393:440–442.

Williams, R. J., and N. D. Martinez. 2004. Limits to Trophic Levels and Omnivory in Complex Food Webs: Theory and Data. The American Naturalist 163:458–468.

Yodzis, P., and K. O. Winemiller. 1999. In Search of Operational Trophospecies in a Tropical Aquatic Food Web. Oikos 87:327–340.

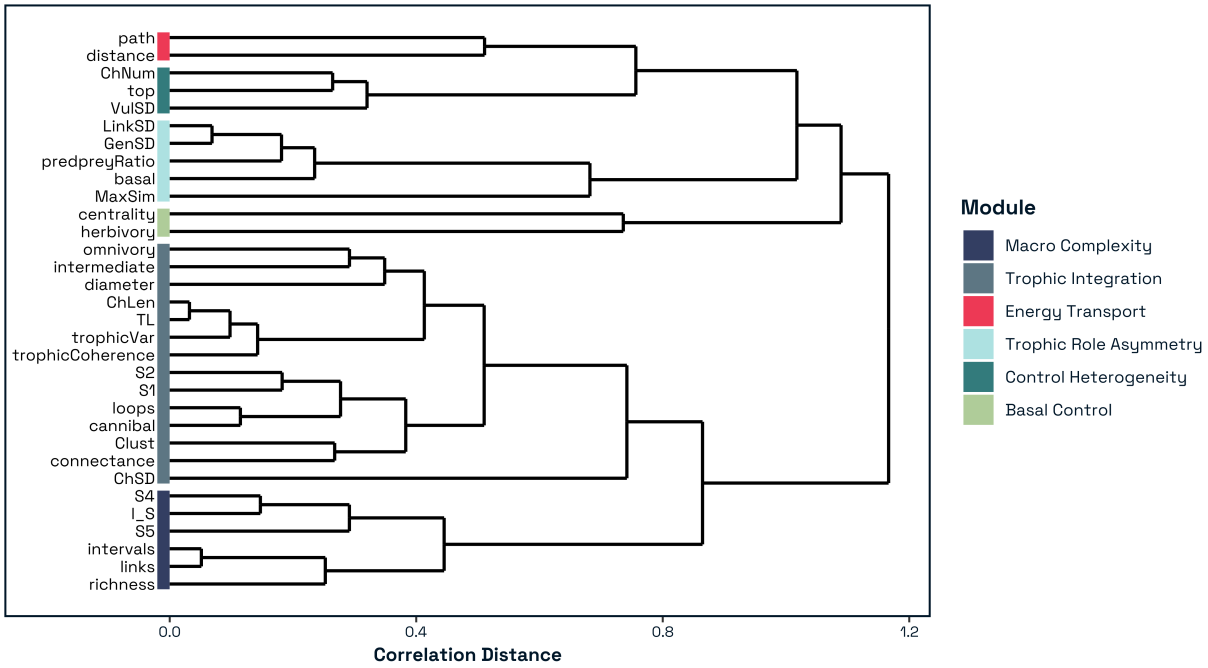

Figure 1: Hierarchical clustering of food-web structural descriptors. Dendrogram illustrating the modular organization of structural metrics based on a signed correlation matrix. Clustering was performed using UPGMA. Coloured bars indicate the seven distinct modules, reflecting robust association patterns among metrics. The horizontal axis represents the correlation distance (1−r), where shorter branches indicate stronger associations between descriptors.

Figure 2: PCA loadings plot illustrating the dominant gradients of food web variation. Each point denotes a structural metric, with colours corresponding to the modular groupings. The clustering of like-coloured points along specific PC axes demonstrates that the modular organisation is preserved within a reduced dimensional space.

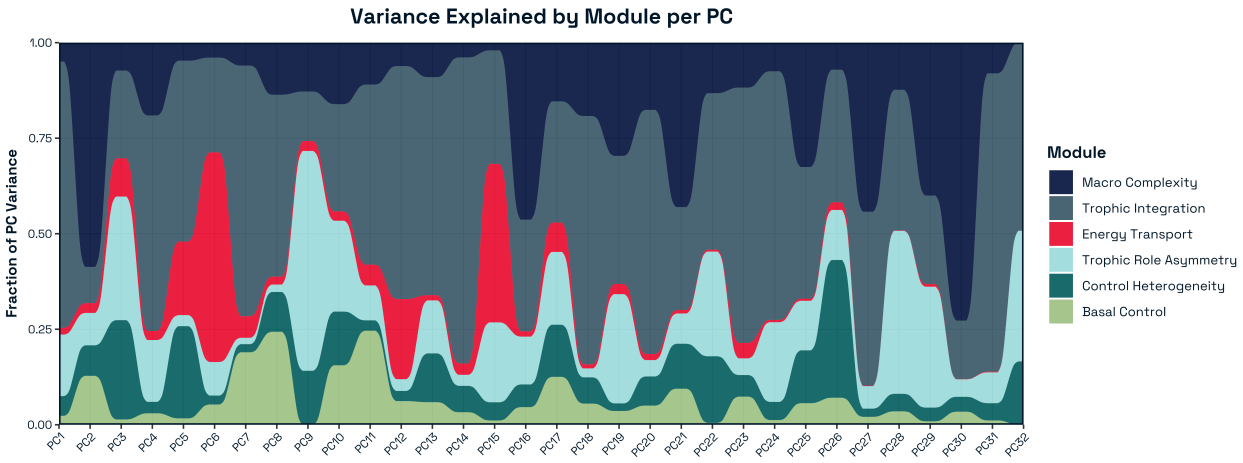

Figure 3: This figure quantifies the relative contribution of each structural module to the variance captured by individual Principal Components (PCs). Calculation: For each PC, the total variance (eigenvalue) is partitioned among the k=7 modules identified in the hierarchical clustering. The contribution of a module to a specific PC is calculated as the sum of the squared loadings of all metrics belonging to that module, normalised by the total variance explained by that PC. Visual Interpretation: The alluvial plot illustrates how the ecological weight shifts across different structural dimensions. For example, while PC1 may be dominated by a single module, subsequent PCs often represent a more diverse blend of modular contributions, reflecting the multifaceted nature of topological complexity and trophic organization. Significance: This visualisation confirms that the PC axes are not merely statistical artifacts but are grounded in the specific groups of correlated metrics defined in the main text. The stability of the flow across the first several components demonstrates that the modular organisation of food web architecture is consistently represented across the primary gradients of structural variation.

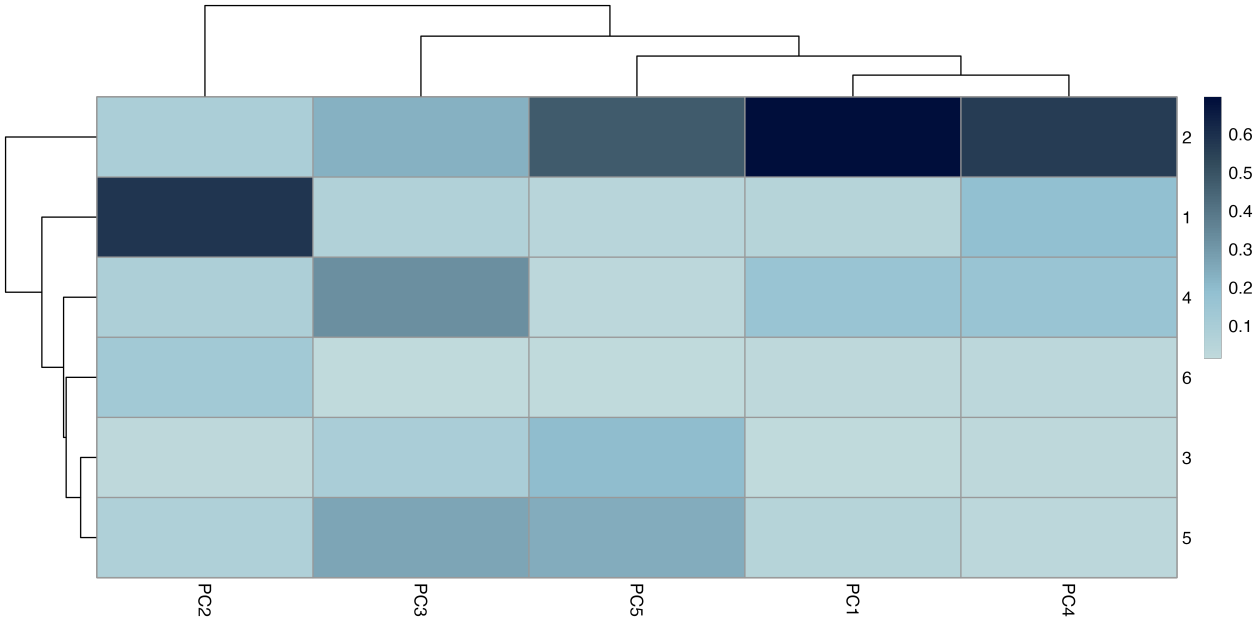

Figure 4: Heatmap showing the correspondence between the seven identified food-web modules (rows) and the principal components (columns) capturing 80% of cumulative variance. High values (darker shading) indicate strong alignment, where a specific module's internal structure drives the variation captured by that PC. Both rows and columns are hierarchically clustered to highlight similarities in how different structural dimensions contribute to the overall food-web architecture.

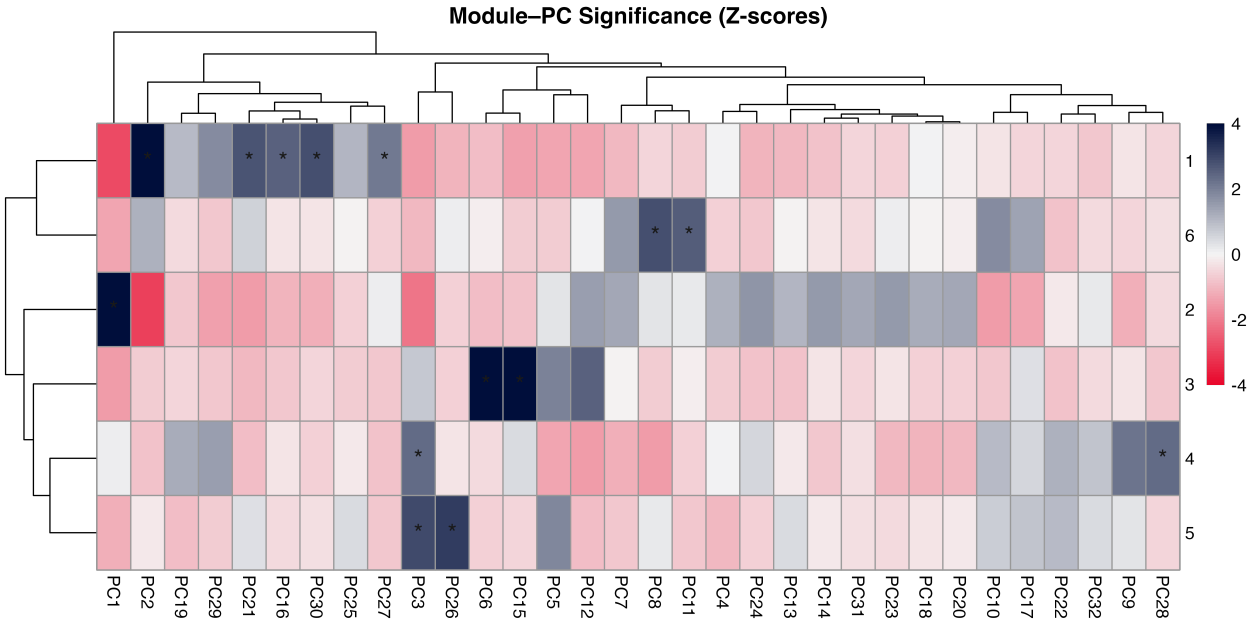

Figure 5: Heatmap showing module–principal component alignment quantified by permutation-derived Z-scores. Colour intensity reflects the magnitude and direction of alignment, and asterisks denote cells that meet joint significance criteria (permutation $P < 0.05$ and $|Z| > 1.96$). Rows correspond to structural metric modules and columns to principal components. Clustering highlights patterns of shared alignment across modules and multivariate structural gradients.

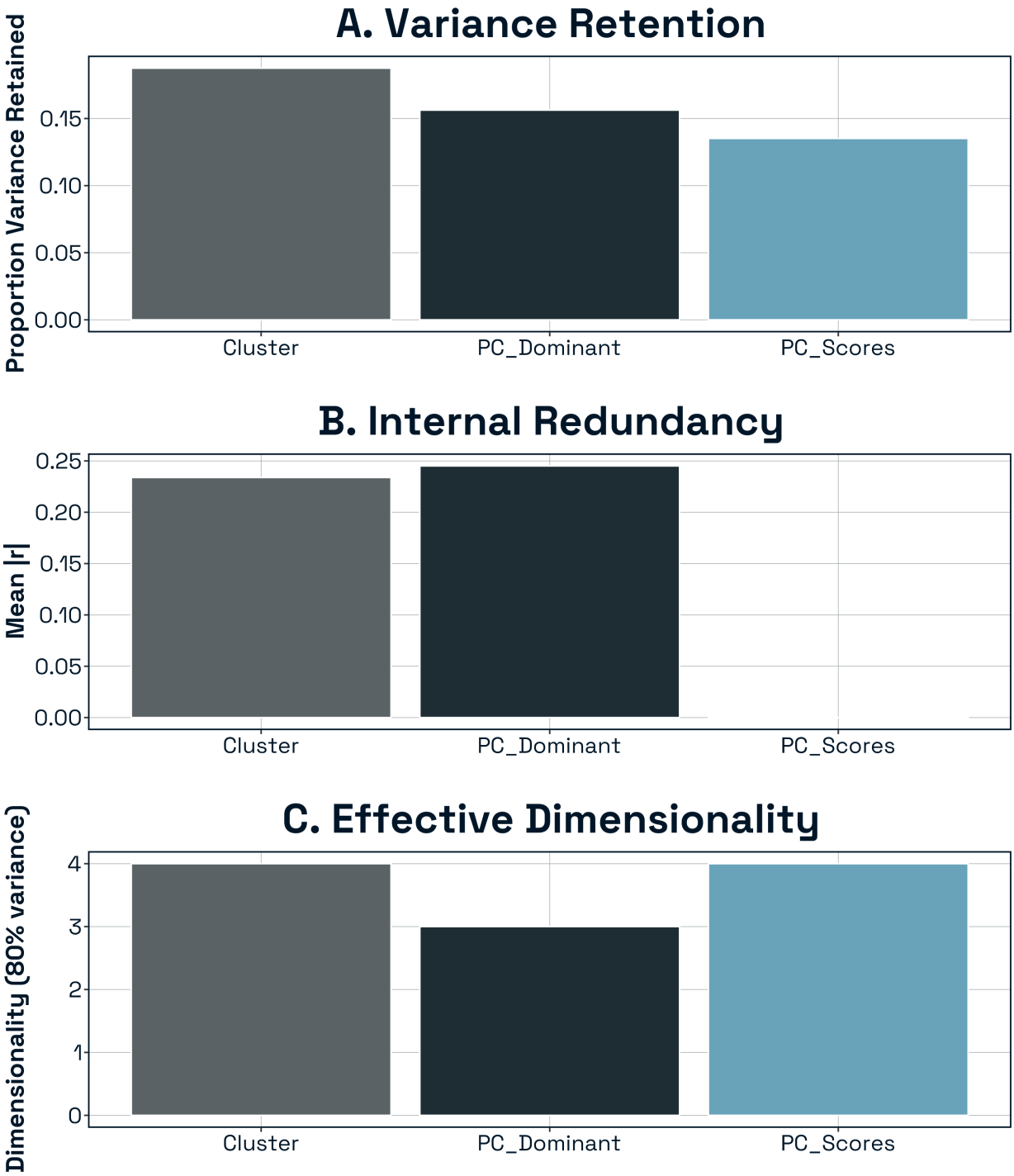

Figure 6: *Figure S1.* Characterisation of alternative dimensional-reduction approaches for network structural metrics. (A) Proportion of total variance in the full metric space retained by each representation: cluster medoids, PC-dominant metrics, and retained PC scores (80% cumulative variance). (B) Mean absolute pairwise correlation (|r|) among predictors within each representation, quantifying internal redundancy. (C) Effective dimensionality, defined as the number of axes required to explain 80% of variance within each reduced predictor set. Together, these diagnostics illustrate differences in information retention, redundancy, and structural alignment among dimensional-reduction strategies.

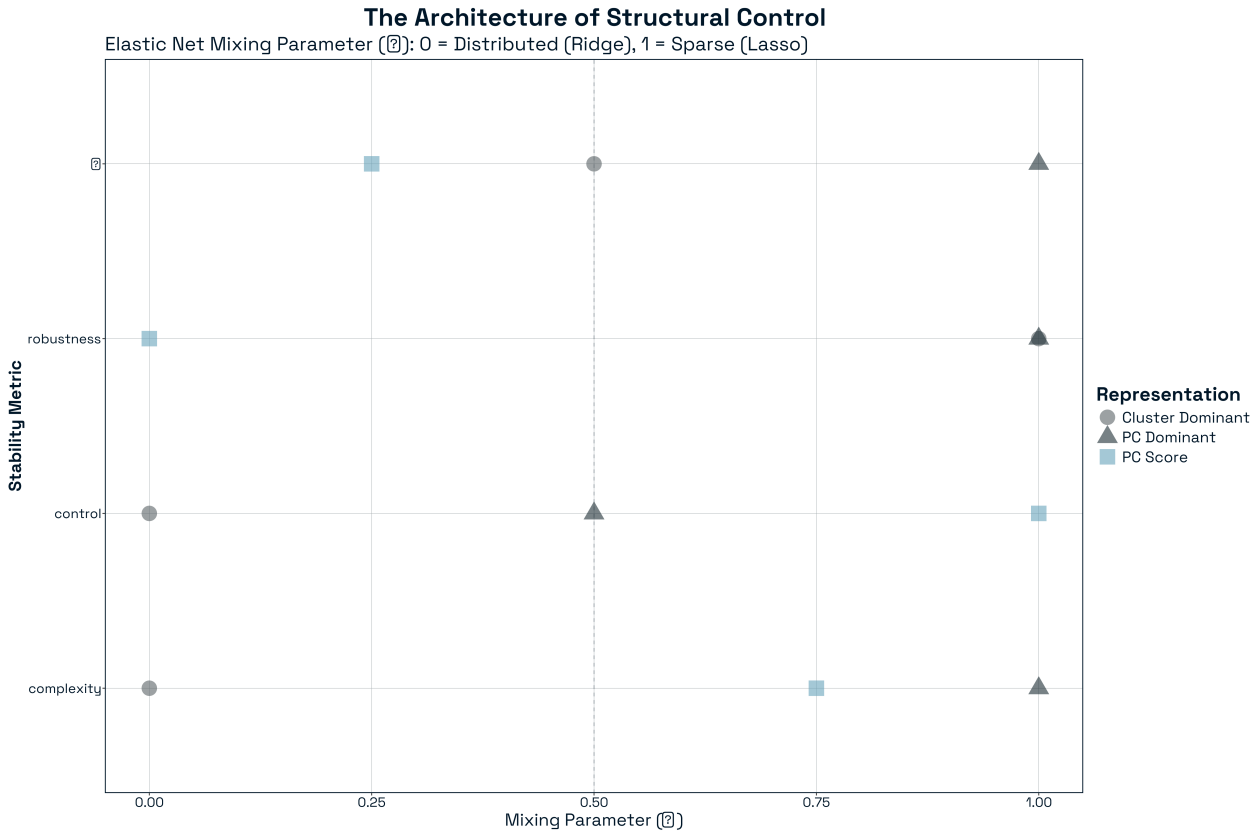

Figure 7: The architecture of structural control across stability regimes. The elastic net mixing parameter ( ) is plotted for each stability metric, categorised by structural representation. values near 0 (Ridge-like) indicate Distributed Control, where stability is an emergent property of the entire network fabric. values near 1 (Lasso-like) indicate Sparse Control, where stability is governed by a small number of specific structural bottlenecks. A dashed line at =0.5 denotes the transition between sparse and distributed modelling regimes.

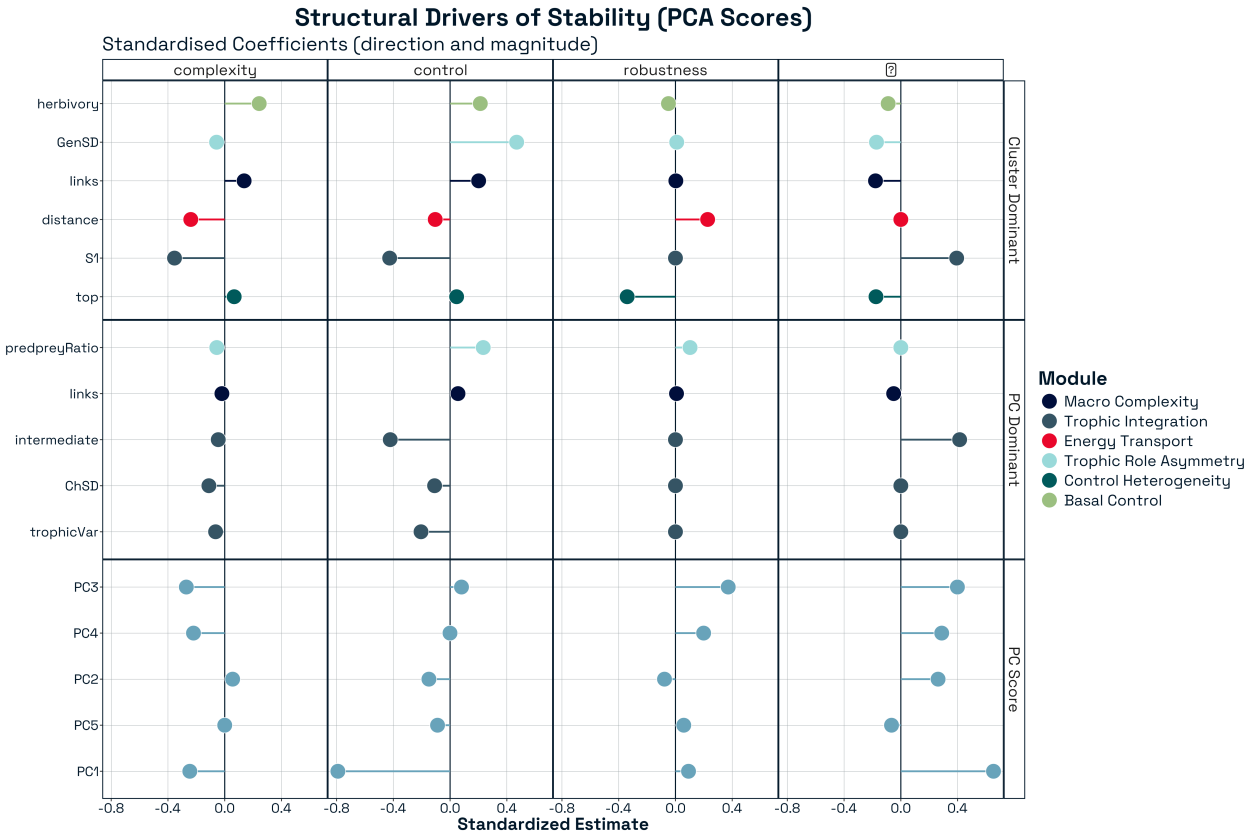

Figure 8: Standardised structural drivers of ecosystem stability. Lollipop plots display the standardised coefficients (estimates) for the top structural predictors across all stability metrics and representations. The magnitude of the point indicates the strength of the effect, while the position relative to the zero-intercept indicates a stabilising (positive) or destabilising (negative) influence. Plots are faceted by representation (rows) and stability metric (columns), with colours identifying the functional module of each structural driver.

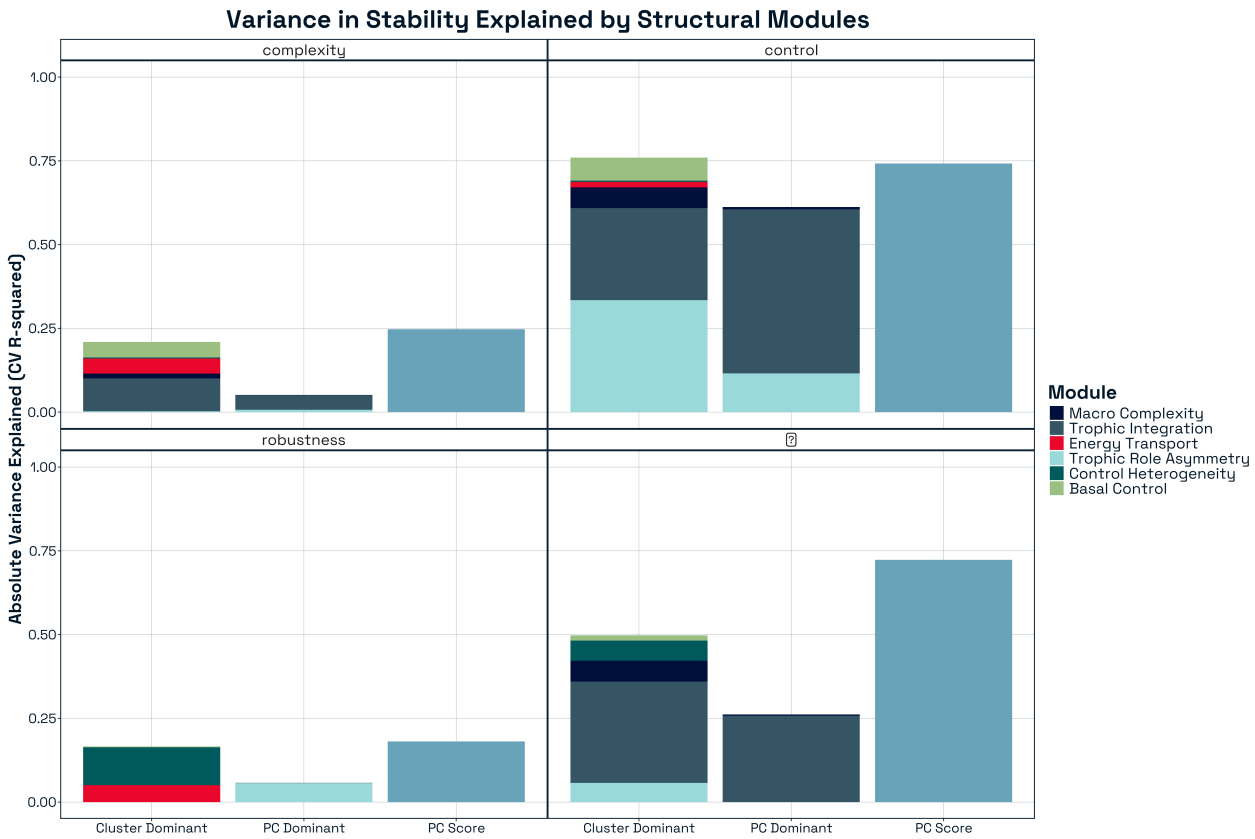

Figure 9: Structural partitioning of variance explained in ecosystem stability. Stacked bar charts represent the absolute variance (cross-validated R2) explained by different structural modules across four stability metrics: robustness, , complexity, and control. Results are faceted by structural representation.

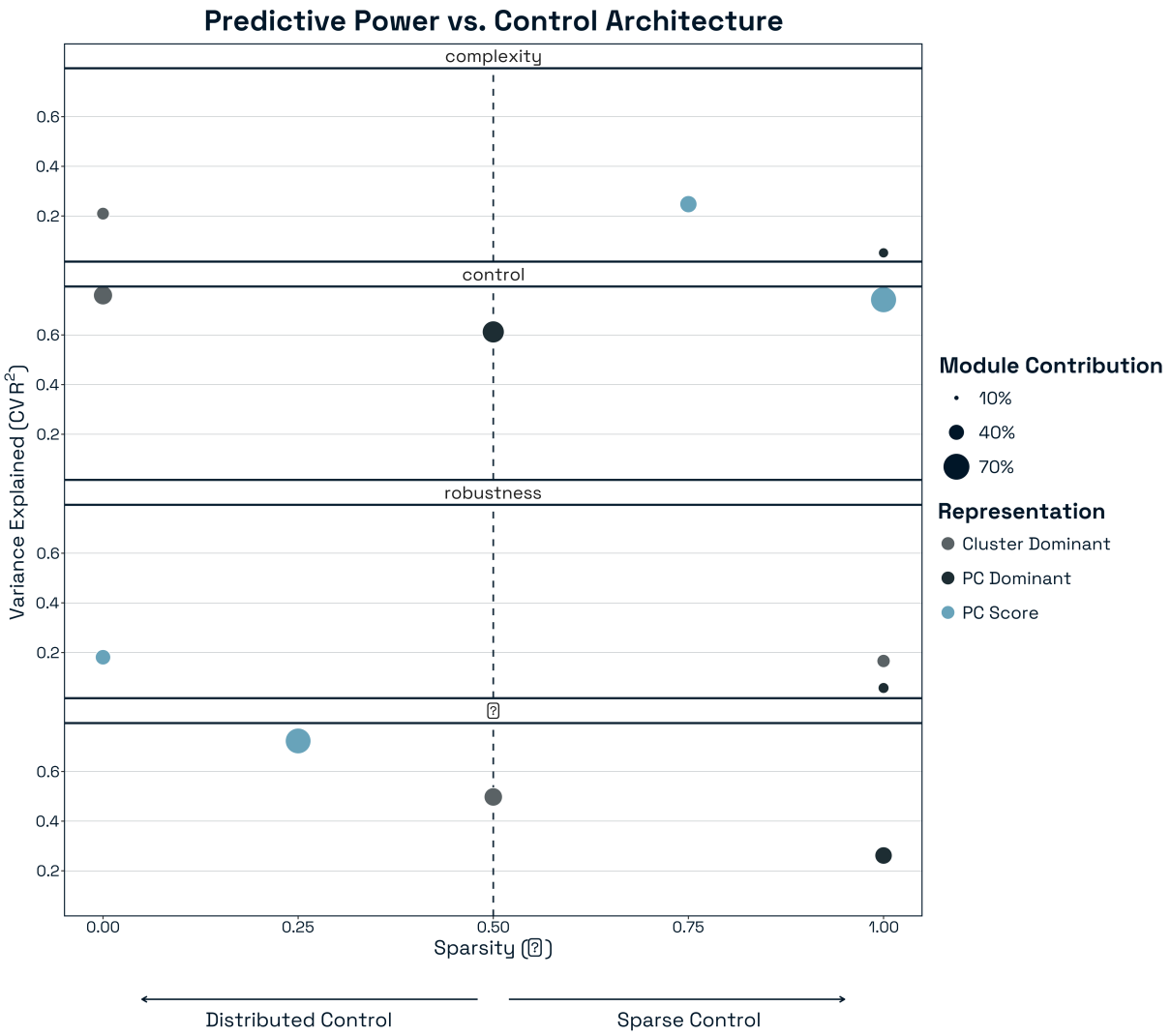

Figure 10: Relationship between predictive power and structural sparsity. Predictive power (CV R2) is plotted against the sparsity index ( ) for all stability metrics. Point size represents the contribution of the dominant structural module to the total variance explained. The horizontal axis identifies the control regime, ranging from Distributed Control (low) to Sparse Control (high). This visualisation reveals whether high predictive accuracy in ecosystem stability is a result of specific structural "levers" (top-right) or systemic, redundant configurations (top-left).